

Working Title

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Abstract

1 Introduction

2 Related Work

3 Methods

We study model performance along three axes: supervision target, inference protocol, and rationale source. Training that supervises only the final score is referred to as score-only supervision (SO), whereas training that supervises both the rationale and final score is referred to as rationale-and-score supervision (RS). Direct-score inference (DS) outputs the score directly, while rationale-first inference (RF) generates a rationale before the score. Using standard RS as the baseline, we further compare three methods: ReaSon, PaIR, and RePaIR.

We first formalize standard RS. Let x_i denote the input of the i -th example, s_i its gold score, and r_i^T the rationale generated by the teacher model. The target completion is

$$y_i^R = \langle \text{reasoning} \rangle r_i^T \langle / \text{reasoning} \rangle \\ \langle \text{score} \rangle s_i \langle / \text{score} \rangle.$$

Standard RS averages token losses over the entire completion. Its objective can therefore be decomposed as

$$\mathcal{L}_{\text{RS}} = \frac{N_r}{N_r + N_s} \mathcal{L}_r + \frac{N_s}{N_r + N_s} \mathcal{L}_s,$$

where N_r and N_s are the numbers of rationale and score tokens, and $\mathcal{L}_r$ and $\mathcal{L}_s$ are their respective mean token-level losses. When $N_r \gg N_s$, the rationale span receives a larger weight in the overall scalar loss.

3.1 ReaSon: Regenerated Rationales with Gold-Score Restoration

ReaSon replaces teacher rationales with student-generated rationales while retaining the standard RS objective. Here, regeneration refers only to rationale regeneration followed by gold-score restoration. It does not involve an explicit critique-revision loop.

Teacher rationales r_i^T are distilled from deepseek-v4-pro and filtered for format validity and gold-score consistency, forming $\mathcal{D}_T = \{(x_i, r_i^T, s_i)\}$. A first-stage model θ_A is trained on $\mathcal{D}_T$ and then generates new rationales r_i^A under RF. Its predicted scores are retained only for recording, while the gold scores remain the training labels. This produces $\mathcal{D}_A = \{(x_i, r_i^A, s_i)\}$.

The second-stage model is reinitialized from the same base model rather than obtained by further updating θ_A . ReaSon therefore primarily replaces r_i^T with r_i^A as the source of rationale supervision. The resulting model is evaluated under both DS and RF. Further details on data construction, prompts, and generation are provided in Appendix A.

3.2 Paired-Interface Reweighting (PaIR)

Standard RS has two limitations: long rationales receive greater weight under token-averaged training, and only the RF interface is observed during training. To address the former, we separately weighted the rationale and score regions within a single RF sequence and prevented the score region from attending to the rationale. However, this design still uses only RF prompts and therefore cannot resolve the DS interface mismatch.

PaIR constructs paired DS and RF training views for each example. The two views share the same user prompt and gold score but differ in their system prompts and output structures. The DS target y_i^D contains only the score block $\langle \text{score} \rangle s_i \langle / \text{score} \rangle$. The RF target y_i^R adds a

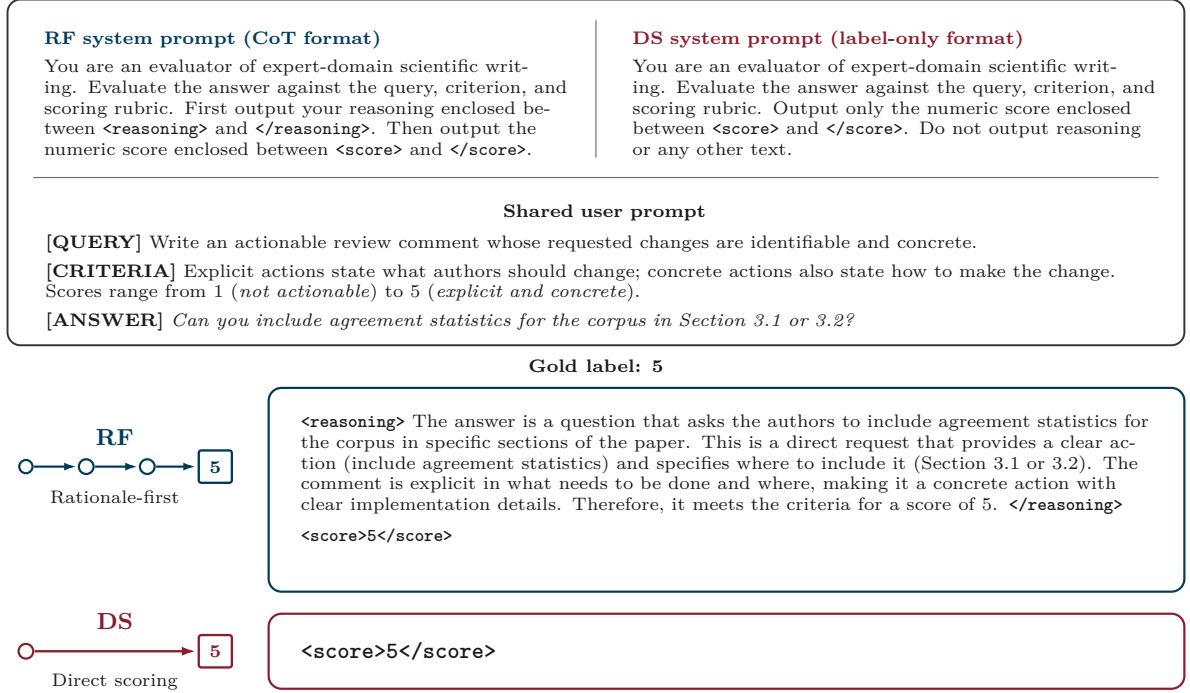


Figure 1: Matched direct-score (DS) and rationale-first (RF) evaluation prompts and outputs for the same scientific-writing example.

rationale block `<reasoning>` r_i `</reasoning>` before the same score block.

The mean token-level losses of the two views are denoted by $\mathcal{L}_D$ and $\mathcal{L}_R$. They are computed separately and combined with equal weight:

$$\mathcal{L}_{\text{PaIR}} = 0.5\mathcal{L}_D + 0.5\mathcal{L}_R.$$

The resulting scalar loss is used for a single backward pass and parameter update. This objective balances the DS and RF views rather than the rationale and score tokens within the RF view.

We use Mix as a matched ablation. It shares the same data and training configuration as PaIR but removes branch-wise normalization. PaIR outperforms Mix under both DS and RF, indicating that branch-wise normalization provides additional gains.

3.3 Regenerated-Rationale Paired-Interface Reweighting (RePaIR)

RePaIR combines the rationale replacement of ReaSon with the paired supervision of PaIR. Starting from the base model θ_0 , the first stage trains a PaIR model θ_A on the teacher-rationale dataset $\mathcal{D}_T$. The resulting model then regenerates rationales r_i^A under RF. These rationales are paired with the gold scores s_i to form $\mathcal{D}_A = \{(x_i, r_i^A, s_i)\}$.

The second stage reinitializes the model from θ_0 and applies PaIR to $\mathcal{D}_A$, rather than further updating θ_A . The resulting model is evaluated under DS and RF, denoted by $\text{RePaIR}_{\text{DS}}$ and $\text{RePaIR}_{\text{RF}}$, respectively.

Comparing ReaSon with RS assesses the effect of rationale source. Comparing PaIR with Mix assesses branch-wise normalization. Comparing RePaIR with PaIR assesses the additional effect of replacing teacher rationales with student-generated rationales.

4 Experiments

Datasets. Our training data were adapted from (Sahinuç et al., 2026)’s work through task selection, prompt revision, and sample-level quality control. The resulting dataset covers two complementary dimensions of scientific-writing evaluation. Peer-review quality is scored on a five-point scale, while related-work assessment is formulated as binary classification. Further details on data processing and dataset statistics are provided in Appendix B.

Models. We used Qwen3-4B (Yang et al., 2025) as the base model for all experiments. For each task–method combination, we fine-tuned a separate model initialized from the same Qwen3-4B checkpoint.

Implementation details. We use LoRA to compare RS and SO supervision for scientific-writing evaluation. All seven datasets share the same training, validation, and test splits. Separate adapters are trained for each task and method, while the validation and test sets are used for model selection, analysis, and final evaluation. All evaluations use zero-shot prompts, with templates shown in Figure 1. LoRA is applied to all linear layers with $r = 16$, $\alpha = 32$, a learning rate of 1×10^{-4} , three epochs, cosine decay, and a maximum sequence length of 8,192 tokens. Additional settings are provided in Appendix A. Evaluating two supervision strategies across seven datasets and two inference protocols yields $2 \times 7 \times 2 = 28$ experimental conditions, reported in Section 5. We then analyze validation-set NLL and rationale quality under RF, followed by sentence-level analysis of prediction changes from DS to RF. For harmful changes, erroneous rationale content is replaced before rescoreing to assess its effect on the prediction. Finally, we evaluate ReaSon and PaIR separately before testing their combination, RePaIR.

4.1 Understanding the SO–RS Performance Gap

To test whether the SO–RS performance gap is associated with teacher-rationale fitting, we compute teacher-forced rationale NLL for both methods across three training seeds (42, 43, and 44). The analysis covers seven tasks: Actionability, Grounding Specificity, Helpfulness, Verifiability, Coherence, Positioning Check, and Positioning Type. For each task, all six adapters use the same validation split. These splits contain 1,368 samples and 161,178 teacher-rationale tokens in total.

For each adapter, complete prompt–rationale–score sequences are processed individually, while loss is computed only at rationale-token positions. Under causal masking, each token conditions only on the prompt and preceding teacher rationale tokens. We then micro-average the loss over all rationale tokens:

$$\text{NLL}_{\text{rat}} = \frac{\sum_n \sum_{t \in R_n} -\log p_\theta(r_{n,t} \mid x_n, r_{n,<t})}{\sum_n |R_n|}.$$

Here, R_n denotes the rationale-token positions in sample n . This yields three seed-level NLL values per method and task, which are averaged for reporting. A lower NLL indicates closer fitting to the teacher-rationale distribution.

Rationale Quality Comparison. Because NLL measures only teacher-rationale fitting, we further use external LLMs to assess generated-rationale quality and consistency with predicted scores. This analysis uses RF outputs from seed 42 and covers the same seven tasks. We randomly sample 50 paired SO–RS cases per task, yielding 350 pairs and 700 rationales.

GLM-5.3-Flash and Doubao-Seed-2.0-Lite serve as primary judges, with MiniMax-M3 as the adjudicator. Adjudication is triggered by invocation or parsing failure, judge disagreement, or a difference of at least two points on any dimension. It is used for 192 of the 350 pairs. The first round scores each rationale from 1 to 3 on rubric understanding, evidential support, key-information coverage, and unsupported-claim control. Mean scores determine whether one rationale is better or both are comparable. The second round classifies whether each rationale supports its predicted score as supported, partially_supported, or not_supported. Model configurations and evaluation prompts are provided in Appendix B.

4.2 Understanding the DS–RF Performance Gap on Ordinal Tasks

Across the 28 experimental conditions, the effect of the inference protocol varied by task type and training objective. DS consistently outperformed RF on ordinal tasks. On binary tasks, DS performed better under SO, whereas RF performed better under RS. We therefore compared sample-level DS-to-RF transitions, together with changes in Accuracy, QWK, and outcome distributions. The analysis covered four ordinal tasks and three random seeds, yielding 11,364 paired DS–RF predictions per training objective. Ordinal predictions were categorized as correct, adjacent errors, severe errors, or format-invalid.

For samples whose predictions changed, we applied the same three-judge protocol. Given the test instance, gold score, DS and RF predictions, and RF rationale, the judges determined whether the rationale supported the gold score, the incorrect prediction, or neither due to insufficient evidence. Harmful transitions were further annotated sentence by sentence for errors such as factual_error and evidence_misread. The complete taxonomy is provided in Appendix A.

Finally, GLM-5.3-Flash corrected the rationales of 158 harmful cases, which were then reviewed by MiniMax-M3. Each case was rescored using

its original adapter and the same RF prompt. Condition A used standard RF inference to generate both the rationale and score. Condition B fixed the original erroneous rationale and regenerated only the score, whereas Condition C fixed the corrected rationale and regenerated only the score. The A–B comparison tested whether fixed-prefix rescoring reproduced the original RF prediction, while B–C measured the prediction change resulting from rationale correction.

5 Results

All primary analyses use strict parsing to evaluate both scoring performance and format compliance, accepting only schema-conforming scores; relaxed-parsing results are reported in Appendix A.

RQ1: How Do SO and RS Differ in Scoring Performance and Rationale Modeling? Figure 2 reports QWK on the four ordinal tasks. With score-only training, direct-score and rationale-first inference achieve mean QWK scores of 0.747 and 0.712, respectively. With rationale-and-score training, the corresponding scores are 0.696 and 0.679. This pattern is consistent across tasks: score-only training outperforms rationale-and-score training, and direct-score inference outperforms rationale-first inference.

Figure 3 reports Macro-F1 on the binary tasks. With score-only training, direct-score and rationale-first inference perform similarly, scoring 0.928 and 0.923, respectively. With rationale-and-score training, rationale-first inference reaches 0.922, compared with 0.843 under direct-score inference. The effect of the training objective therefore depends on the inference protocol, with a clear score-only advantage appearing only under direct-score inference.

Rationale supervision improves teacher-rationale fitting, but this advantage does not translate into better scoring. We measure teacher-rationale modeling using teacher-forced rationale NLL. The original Qwen3-4B without task-specific fine-tuning (Base) obtains a mean NLL of 3.4480, which decreases to 1.6964 with score-only training and 0.7793 with rationale-and-score training. Rationale-and-score training achieves the lowest NLL on every task, showing that direct rationale supervision improves teacher-token prediction but does not guarantee better scoring performance.

Rationale-and-score training produces higher-quality rationales. External LLMs compared 350

Task	Base	SO	RS	RS – SO
Actionability	3.3984	1.5891	0.7894	−0.7997
Grounding Specificity	3.6439	1.6757	0.6761	−0.9996
Helpfulness	3.5397	1.7811	0.8856	−0.8955
Verifiability	3.5677	1.9243	0.8401	−1.0842
Coherence	3.4272	1.4634	0.7852	−0.6782
Positioning Check	3.1925	1.5362	0.7112	−0.8250
Positioning Type	3.3664	1.9051	0.7676	−1.1375
Average	3.4480	1.6964	0.7793	−0.9171

Table 1: Teacher-forced rationale NLL by task; lower is better.

paired rationales and preferred those from rationale-and-score training in 248 cases (70.9%). Score-only rationales were preferred in 25 cases (7.1%), while 77 cases (22.0%) were comparable. This preference was consistent across all evaluation tasks. Rationale-and-score training scored higher on rubric understanding, evidence grounding, key-information coverage, and unsupported-claim control, with the largest difference in key-information coverage. Its mean score across the four dimensions was 2.832, compared with 2.400 for score-only training, and its rationale–score consistency was also higher at 96.6% versus 74.3%.

RQ2: Why Does RF Underperform DS on Ordinal Tasks? We traced sample-level DS-to-RF transitions across four ordinal tasks and three seeds, yielding 11,364 paired events per supervision objective. Switching to RF reduced mean QWK by 0.035 under SO and 0.017 under RS. Under SO, accuracy decreased by 3.52 percentage points, while severe errors increased by 3.98 points. Under RS, accuracy rose by 3.49 points, yet severe errors increased by 4.26 points. Because QWK penalizes cross-level errors more heavily, this increase offset the gain in exact predictions. Invalid-output rates changed by less than 0.25 points and were not the main source of the QWK decline.

To localize these severe errors, we audited 296 harmful or beneficial transition cases, of which 278 reached valid judge consensus. The judges found that 158 RF rationales supported an incorrect score, whereas 120 supported the gold score. Rubric misapplication was the most frequent sentence-level error, followed by score-mapping errors, evidence misreadings, and factual errors. The first error appeared near the second sentence on average, and about 69% occurred within the first two sentences. These observations linked rationale direction to prediction transitions, but did not establish causality.

Analysis	Measure	SO	RS	Comparison
Pairwise Preference	Preferred rationale, n (%)	25 (7.1)	248 (70.9)	Tie: 77 (22.0)
Quality	Rubric understanding	2.379	2.848	+0.469
	Evidence grounding	2.495	2.837	+0.342
	Key-information coverage	2.273	2.833	+0.560
	Unsupported-claim control	2.453	2.811	+0.358
	Four-dimension mean	2.400	2.832	+0.433
Consistency	Supported (%)	74.3	96.6	+22.3 pp
	Partially supported (%)	20.0	3.1	−16.9 pp
	Not supported (%)	5.7	0.3	−5.4 pp

Table 2: External-LLM evaluation of 350 paired rationales generated under rationale-first inference. Quality dimensions use a 1–3 scale. For pairwise preference, Comparison reports tied judgments; elsewhere, it reports RS minus SO, with percentage differences in percentage points.

We therefore applied a fixed-prefix intervention to the 158 cases whose rationales supported an incorrect score. Condition A used the natural RF output. Condition B fixed the original erroneous rationale, whereas Condition C fixed its corrected counterpart before regenerating the score. Condition B reproduced 157 of 158 original predictions, showing that prefix fixation rarely changed the score by itself. Replacing B with C changed every prediction and restored 142 cases (89.87%) to gold. The remaining 16 became adjacent errors, while MAE fell from 2.044 to 0.101. This intervention establishes sensitivity to the exposed rationale on the selected cases, not complete faithfulness of the model’s hidden computations.

5.1 RQ3: How Can RS Preserve Rationale Quality and Scoring Performance?

ReaSon first changed the rationale source used along the RF path. It increased mean ordinal QWK from 0.679 to 0.707 and reduced MAE from 0.583 to 0.534. Classification Macro-F1 remained nearly unchanged, raising the seven-task mean from 0.783 to 0.799. ReaSon also reduced the error in 172 of 338 original RS harmful transitions (50.9%). However, its relaxed DS mean changed by only -0.001 , while its strict invalid-output rate reached 47.74%. The gains were therefore concentrated under RF and did not resolve DS interface coverage.

Reweighting a single RF sequence did not solve this problem. Across 34 completed evaluations, DS produced 2,387 strict-format failures, although relaxed parsing recovered 2,378 of them. PaIR instead paired DS and RF training views to provide explicit interface coverage. Its seven-task DS mean reached 0.833, exceeding SO_DS (0.825) and RS_DS (0.768). Its RF mean reached 0.797,

exceeding RS_RF (0.783) but remaining below SO_RF (0.803). Against Mix, which used the same paired data, PaIR gained 0.013 under DS and 0.006 under RF. This matched ablation supports an additional benefit from branch-wise normalization.

Finally, RePaIR combined ReaSon’s rationale regeneration with PaIR’s paired-interface supervision. RePaIR_DS achieved a seven-task mean of 0.832, which was 0.007 above SO_DS and 0.001 below PaIR_DS. RePaIR_RF reached 0.817, exceeding SO_RF by 0.014 and PaIR_RF by 0.020. Both interfaces maintained 100% strict-format validity. RePaIR therefore preserved nearly all of PaIR’s DS gain while adding the RF benefit of rationale regeneration, yielding the most balanced aggregate performance across the two interfaces. These results verify the scoring and formatting gains. A matched audit is still needed to test whether RePaIR retains ReaSon’s rationale-quality improvement.

References

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